

International Institute of Information Technology, Bhubaneswar

Department of Computer Science and Engineering

Object Oriented Programming Laboratory**Lab 2 – C++ Programming: Classes & Objects****Date:** 04.08.2026**Semester:** B.Tech 3rd Semester**Section:** CSE B2**Instructions**

- Write all programs in C++.
- Use **classes and objects** in every program.
- Implement the required functionality using **member functions**.
- Display the output in a neat and readable format.
- Use appropriate data types and meaningful variable names.

1. Student Information System

Create a class named `Student` to store the details of a student. The class should contain the following data members:

- Roll Number
- Name
- Marks obtained in one subject

Write suitable member functions to:

1. Accept the student's details from the user.
2. Display the entered details in a formatted manner.

***Hint:** Create an object of the class in the `main()` function. Use one function for input and another for displaying the information.*

2. Rectangle Calculator

Design a class named `Rectangle` that stores the **length** and **breadth** of a rectangle. Write member functions to:

1. Read the dimensions from the user.
2. Calculate the area of the rectangle.
3. Calculate the perimeter of the rectangle.
4. Display all the results.

***Hint:** $\text{Area} = \text{Length} \times \text{Breadth}$, and $\text{Perimeter} = 2 \times (\text{Length} + \text{Breadth})$.*

3. Simple Calculator

Create a class named `Calculator` that performs basic arithmetic operations on two numbers. Implement separate member functions to perform:

- Addition
- Subtraction
- Multiplication
- Division

Display the result of each operation.

Hint: Check that the second number is not zero before performing division.

4. Bank Account Management

Design a class named `BankAccount` to manage a customer's bank account. The class should contain:

- Account Number
- Account Holder Name
- Balance

Write member functions to:

1. Enter account details.
2. Deposit money into the account.
3. Withdraw money from the account.
4. Display the updated account details.

Hint: Do not allow withdrawal if the requested amount is greater than the available balance. Display an appropriate message when withdrawal is unsuccessful.

5. Employee Salary Calculator

Create a class named `Employee` to maintain employee information. The class should include:

- Employee ID
- Employee Name
- Basic Salary

Write member functions to:

1. Accept employee details.
2. Calculate: $\text{HRA} = 20\% \text{ of Basic Salary}$, $\text{DA} = 10\% \text{ of Basic Salary}$, and $\text{Gross Salary} = \text{Basic Salary} + \text{HRA} + \text{DA}$.
3. Display all salary details.

Hint: Store HRA, DA, and Gross Salary as separate variables before displaying them.

6. Distance Addition

Create a class named `Distance` with two data members: **Feet** and **Inches**. Write member functions to:

1. Input two distances.
2. Add the two distances.
3. Display the final distance.

***Hint:** If the total inches are 12 or more, convert them into feet. Example: 5 ft 10 in + 3 ft 8 in = 9 ft 6 in.*

7. Product Inventory Management

Design a class named `Product`. The class should contain:

- Product ID
- Product Name
- Quantity Available
- Price per Unit

Write member functions to:

1. Accept product details.
2. Display the product details.
3. Update the quantity after selling a specified number of units.
4. Calculate and display the total inventory value.

***Hint:** $\text{Inventory Value} = \text{Quantity} \times \text{Price}$. Ensure that the quantity sold does not exceed the available stock.*

8. Library Book Management System

Create a class named `LibraryBook` to manage the issue and return of books. The class should contain:

- Book ID
- Book Title
- Student Name
- Number of Days the Book was Issued

Write member functions to:

1. Enter the book and student details.
2. Calculate the fine for late return.
3. Display all transaction details.

Fine Rule: No fine for the first 15 days. ₹2 per day for each additional day.

***Hint:** $\text{Fine} = (\text{Days Issued} - 15) \times ₹2$, only if $\text{Days Issued} > 15$.*

9. Student Result Processing System

Create a class named `StudentResult`. The class should contain:

- Student Name
- Roll Number
- Marks in Five Subjects

Write member functions to:

1. Accept the student's details.
2. Calculate the Total Marks and Percentage.
3. Assign grades according to the following criteria and display the complete result:

Percentage	Grade
90% and above	A
80–89%	B
70–79%	C
60–69%	D
Below 60%	F

Hint: Percentage = (Total Marks / 500) × 100. Assume each subject carries 100 marks.

10. Electricity Bill Generator

Create a class named `ElectricityBill`. The class should contain:

- Consumer Number
- Consumer Name
- Units Consumed

Write member functions to accept the consumer details, calculate the electricity bill using the following slab rates, and display the complete bill showing consumer details, units consumed, and total bill amount.

Units	Rate
First 100 units	₹5 per unit
Next 100 units	₹7 per unit
Above 200 units	₹10 per unit

Hint: Calculate the bill slab-wise. Example for 250 units: $(100 \times 5) + (100 \times 7) + (50 \times 10) = ₹1700$.